

OS PRACTICAL

PRACTICAL3

Develop a C program using fork() that creates a parent and child process. Display the Process ID (PID), Parent Process ID (PPID), and process states at different stages of execution.

Design an experiment to observe process state transitions (Ready, Running, Waiting, Terminated) using Linux monitoring tools such as ps, top, and /proc. Document the observations.

OUTPUT:

```
Rashmika@LAPTOP-UNQ0CNLA:~$ cd OS_LAB
Rashmika@LAPTOP-UNQ0CNLA:~/OS_LAB$ cd Practical
Rashmika@LAPTOP-UNQ0CNLA:~/OS_LAB/Practical$ cd Practical3
Rashmika@LAPTOP-UNQ0CNLA:~/OS_LAB/Practical/Practical3$ nano Practical3.c
Rashmika@LAPTOP-UNQ0CNLA:~/OS_LAB/Practical/Practical3$ gcc Practical3.c -o Practical3
Rashmika@LAPTOP-UNQ0CNLA:~/OS_LAB/Practical/Practical3$ ./Practical3

----- Parent Process -----
Parent PID : 1683
Child PID  : 1684
Parent waiting for child...

----- Child Process -----
Child PID  : 1684
Parent PID : 1683
Child is sleeping for 30 seconds...
Child process completed.
Child terminated.
Parent process completed.
Rashmika@LAPTOP-UNQ0CNLA:~/OS_LAB/Practical/Practical3$ ps -ef | grep Practi
cal3
Rashmika    1697    1599  0 15:25 pts/0    00:00:00 grep --color=auto Practi
cal3
Rashmika@LAPTOP-UNQ0CNLA:~/OS_LAB/Practical/Practical3$ top
top - 15:28:26 up 1:20, 1 user, load average: 0.00, 0.00, 0.00
Tasks: 24 total, 1 running, 23 sleeping, 0 stopped, 0 zombie
%Cpu(s):  0.0 us,  0.0 sy,  0.0 ni,100.0 id,  0.0 wa,  0.0 hi,  0.0 si,  0.
MiB Mem :  7762.6 total,  7085.2 free,   533.7 used,   295.0 buff/cache
MiB Swap:  2048.0 total,  2048.0 free,    0.0 used.  7229.0 avail Mem

  PID USER      PR  NI   VIRT   RES   SHR  S  %CPU  %MEM     TIME+
218  _chrony    20   0  12096   2468  1824  S   0.0   0.0   0:00.00
235  root       20   0 123460  32772 18160  S   0.0   0.4   0:00.21
313  root       20   0   8644   5604  4764  S   0.0   0.1   0:00.01
356  Rashmika   20   0  22344  13604 10980  S   0.0   0.2   0:00.08
358  Rashmika   20   0  22916   4092  2088  S   0.0   0.1   0:00.00
386  Rashmika   20   0   6136   5420  3864  S   0.0   0.1   0:00.01
1592 root        20   0   3184   1120   980  S   0.0   0.0   0:00.00
1593 root        20   0   3200   1260  1108  S   0.0   0.0   0:00.04
1599 Rashmika    20   0   6268   5536  3868  S   0.0   0.1   0:00.07
```

```
Rashmika@LAPTOP-UNQ0CNLA:~/OS_LAB/Practical/Practical3$ cat /proc/1599/status
Name:      bash
Umask:     0022
State:     S (sleeping)
Tgid:      1599
Ngid:      0
Pid:       1599
PPid:      1593
TracerPid: 0
Uid:       1001    1001    1001    1001
Gid:       1001    1001    1001    1001
FDSize:    256
Groups:    27 100 1001
NStgid:    1599
NSpid:     1599
NSpgid:    1599
NSSid:     1599
Kthread:   0
VmPeak:    6268 kB
VmSize:    6268 kB
VmLck:     0 kB
VmPin:     0 kB
VmHWM:     5536 kB
VmRSS:     5536 kB
RssAnon:   1668 kB
RssFile:   3868 kB
RssShmem:  0 kB
VmData:    1628 kB
VmStk:     132 kB
VmExe:     1036 kB
VmLib:     1908 kB
VmPTE:     52 kB
VmSwap:    0 kB
HugetlbPages: 0 kB
CoreDumping: 0
THP_enabled: 1
untag_mask: 0xffffffffffffffff
Threads:   1
SigQ:      0/31007
SigPnd:    0000000000000000
```

```
ShdPnd: 000000000000000000  
SigBlk: 00000000000010000  
SigIgn: 0000000000384004  
SigCgt: 000000004b813efb  
CapInh: 0000000000000000  
CapPrm: 0000000000000000  
CapEff: 0000000000000000  
CapBnd: 000001fffffffffffff  
CapAmb: 0000000000000000  
NoNewPrivs:      0  
Seccomp:          0  
Seccomp_filters:    0  
Speculation_Store_Bypass: thread vulnerable  
SpeculationIndirectBranch: conditional enabled  
Cpus_allowed:     fff  
Cpus_allowed_list:   0-11  
Mems_allowed:       00000000,00000000,00000000,00000000,00000000,00000000,00000000,  
00,00000000,00000000,00000000,00000000,00000000,00000000,00000000,00000000,0  
00000000,00000000,00000000,00000000,00000000,00000000,00000000,00000000,00000  
000,00000000,00000000,00000000,00000000,00000000,00000000,00000000,00000001  
Mems_allowed_list:   0  
voluntary_ctxt_switches: 493  
nonvoluntary_ctxt_switches: 0
```